01 - Represent a value with bits

Read, predict and model

Name _____ Class _____ Date _____

Convert between unsigned binary, decimal and hexadecimal with a declared width.

Textbook: representation

Before starting: Use as general supporting computer-systems practice alongside the representation chapter. These activities build conceptual fluency; the CPU and performance examples are supporting context, not a claim that every detail is examinable. Teach binary place value before logic and performance.

For an unsigned 8-bit value, the place values are 128,64,32,16,8,4,2,1. 00101101 equals 32 + 8 + 4 + 1 = 45. Split it into nibbles: 0010 is 2 and 1101 is D, so the hexadecimal representation is 2D. These are representations of the same value.

Eight unsigned bits represent 0 through 255. A calculated value of 256 needs nine bits. The storage width and signed/unsigned interpretation must be stated before interpreting a bit pattern.

Equipment-free route

Use eight labelled place-value cards and counters. Group four bits at a time for hexadecimal; calculators and hardware are optional.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

01 - Represent a value with bits

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent conversions

Convert decimal 58 to 8-bit binary and hexadecimal. Convert binary 10100110 to decimal and hexadecimal. Show place-value working rather than only the final answer.

B. Boundary reasoning

A field stores an unsigned 8-bit count. Can it represent 255, 256 and -1? Explain. Do not assume wraparound behaviour unless the system specification states it.

Value	Fits?	Reason
255		
256		
-1		

Exit

Does leading zero padding change the unsigned numeric value? Why state the width anyway?

02 - Separate characters from encoded bytes

Read, predict and model

Name _____ Class _____ Date _____

Distinguish a code point, encoding and the number of stored bytes.

Textbook: representation

Unicode gives characters code points. For A, U+0041 is decimal 65 and its UTF-8 representation is one byte, hexadecimal 41. For é (precomposed U+00E9), UTF-8 uses two bytes C3 A9. A code point number is not necessarily a byte value in the chosen encoding.

Text	Code points	UTF-8 bytes (hex)
A	U+0041	41
B	U+0042	42
é (precomposed)	U+00E9	C3 A9

UTF-8 uses one to four bytes per code point. Visually similar text may use different code-point sequences. This exercise uses exactly the supplied precomposed character.

Equipment-free route

Use the supplied encoding table as the complete data source. Concatenate byte cards on paper; no Unicode software or VR headset is required.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

02 - Separate characters from encoded bytes

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent byte count

Using the supplied table, write the UTF-8 bytes for ABé and count code points, bytes and bits. Count only encoded text: no byte-order mark, terminator or file metadata.

B. Diagnose an assumption

A learner says ten characters always take ten bytes. Give a counterexample using only the table. Explain why counting displayed symbols can be more complex than counting code points.

Exit

Explain why reading C3 A9 as two unrelated characters would produce the wrong text.

03 - Combine logic and evaluate a small system

Read, predict and model

Name _____ Class _____ Date _____

Construct a truth table and evaluate a declared access decision.

Textbook: selection

A fictional desktop simulation displays OPEN only when staff is true AND (card_valid OR override) is true. The override bypasses the card requirement, not the staff requirement. This is a supplied Boolean rule for reasoning, not a real access-control specification.

```
allowed = staff and (card_valid or override)
```

Staff	Card valid	Override	Allowed
False	False	False	False
False	True	True	False
True	False	True	True
True	False	False	False

Work inside the parentheses first, then apply AND. The output message does not actuate any door or other device.

Equipment-free route

Use Boolean input cards and a paper AND/OR circuit. Trace the declared rule; no electronics or physical access equipment is needed.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

03 - Combine logic and evaluate a small system

Independent practice and evidence

Name _____ Class _____ Date _____

A. Complete the truth table

Build all eight distinct combinations of the three Boolean inputs and calculate allowed. Mark which two rows differ when override changes while staff=True and card_valid=False.

B. Independent comparison

Compare staff and (card_valid or override) with (staff and card_valid) or override. Supply an input combination on which they differ and explain the consequence.

Exit

Which input combinations make the original rule true? Explain without saying merely that the table says so.

04 - Explain processing and a performance bottleneck

Read, predict and model

Name _____ Class _____ Date _____

Use a simplified processing model and distinguish capacity from transfer rate.

Textbook: representation

Supporting context: in a simplified stored-program model, instructions and data reside in memory. The processor fetches an instruction, decodes it and executes its operation; registers hold immediate working values, while persistent storage retains files without power. Real processors overlap work, so this is a conceptual model, not a timing simulator.

A fictional capture system produces 6 MB each second. A storage path sustains 4 MB/s. The backlog grows by 2 MB/s. With an initially empty 20 MB buffer, it fills after 20 / 2 = 10 seconds, assuming constant rates and no other overhead.

Equipment-free route

Draw the pipeline and use counters for produced, written and buffered MB each second. Optional drone/VR recordings can supply later data, but the supplied figures meet the same learning goal.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

04 - Explain processing and a performance bottleneck

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent pipeline

Draw input -> processing -> output for a supplied temperature display. Locate its saved program, working values and saved log using memory/register/storage roles. State which parts must persist after power is removed.

B. Changed performance case

A simulated recorder produces 9 MB/s, writes 6 MB/s and has an empty 24 MB buffer. Calculate net growth and time to fill. Would doubling CPU clock rate necessarily remove this bottleneck? Explain using the rates.

Exit

Distinguish MB of buffer capacity from MB/s of write throughput. Explain one limitation of this constant-rate model.